

PRODUCTION OF STYRENE BY DEHYDROGENATION OF ETHYLBENZENE

CAPACITY: 250 TONS / DAY

**Project Report Submitted in Partial Fulfillment of the
Requirement for the Degree of B.E. Chemical 2025**



by

Priyal Chirania

CH21152

**Dr S.S. BHATNAGAR UNIVERSITY INSTITUTE OF
CHEMICAL ENGINEERING AND TECHNOLOGY,
PANJAB UNIVERSITY, CHANDIGARH-160014, INDIA**

UNDERTAKING

I Priyal Chirania CH21152, of BE Chemical final year certify that present work titled **“PRODUCTION OF STYRENE BY DEHYDROGENATION OF ETHYLBENZENE”** represents solely by my own efforts.

Signature (with Date)

Priyal Chirania

UICET

CERTIFICATE

This is to certify that Priyal Chirania, CH21152, fourth year student of Dr. S.S. Bhatnagar University Institute of Chemical Engineering & Technology, Panjab University, Chandigarh has completed her project titled **“PRODUCTION OF STYRENE BY DEHYDROGENATION OF ETHYLBENZENE”**. This Project Report may be accepted for evaluation as part of the requirement of B.E. (CHEMICAL) Degree.

Signature (with Date)

Mr. Sher Singh

ACKNOWLEDGMENT

I would like to convey my sincere gratitude to my Project Supervisor, Mr. Sher Singh for his invaluable guidance, motivation and suggestions throughout the course of the Project. His encouragement towards the current topic helped me a lot in this Project work which also created an area of interest for my professional career ahead. I own my thankfulness to Dr. Anupama Sharma, Head of the Department for providing necessary facilities and guidance.

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CHAPTER 1

INTRODUCTION

Styrene is primarily a synthetic chemical. It is also known as vinylbenzene, cinnamene or phenylethylene. It is a colourless liquid that evaporates easily and has a sweet smell. It often contains other chemicals that give it a sharp, unpleasant smell. It dissolves in some liquids but doesn't dissolve easily in water.

Styrene is largely produced through the dehydrogenation of ethylbenzene, a process that is energy intensive and typically carried out in the presence of catalysts at high temperatures. Due to its reactive vinyl group, styrene readily undergoes polymerization, forming materials with a wide range of applications, from packaging and insulation to automotive and construction products. It is mostly used in manufacturing of polystyrene, acrylonitrile-butadiene-styrene (ABS), styrene acrylonitrile (SAN), styrene-butadiene rubber (SBR), and lattices, unsaturated polyester resins, and miscellaneous uses like textile auxiliaries, pigment binders polyester resin, aromatics and intermediate industries.

PHYSICAL PROPERTIES

Chemical name	Styrene
Molecular formula	C_8H_8
Molar mass	104.15g/mol
Appearance	Colourless to yellowish oily liquid
Boiling point	145°C
Melting point	-30°C
Density (20°C)	909kg/m ³
Solubility in water	0.3g/L
Vapour pressure (20°C)	6.3 mmHg
Refractive index	1.546
Odor	Sweet, aromatic

APPLICATIONS

Applications of styrene are-

- Styrene is mainly used as raw material for polystyrene, synthetic rubber, plastics, ion exchange resins, etc.
- The most important use of styrene is as monomer for synthetic rubber and plastics, it is used to produce styrene-butadiene rubber, polystyrene, polystyrene foam; it is also used as other co-polymerizable monomers to manufacture many different applications of engineering plastics.
- Styrene is used for the preparation of copper brightener, and has effect of levelling and bright.
- Styrene is used in table food, cake food, condiments, desserts, snacks, all kinds of canned food, candy. Variety of drinks, especially yogurt, lactic acid bacteria drinks, carbonated drinks and other acidic beverages.
- Styrene is used for electron microscopic analysis, organic synthesis.

CHAPTER 2

MANUFACTURING PROCESS AND SELECTION

Many different techniques have been investigated for the manufacture of styrene monomer. Of these, the following methods have been used or seriously considered for commercial production:

1. Dehydrogenation of ethylbenzene
2. Oxidation of ethylbenzene to ethyl benzene hydroperoxide, which reacts with propylene oxide, after which the alcohol is dehydrated to styrene
3. Side chain chlorination of ethylbenzene followed by dichlorination
4. Side chain chlorination of ethyl benzene hydrolysis to the corresponding alcohols followed by dehydration.
5. Pyrolysis of petroleum recovery from various petroleum processes.

Selected method is **Catalytic Dehydrogenation of ethyl benzene** because process reaction is equilibrium limited and with the addition of steam the process can be controlled moreover steam (used to vaporize ethyl benzene) is recycled. So, this process is economical.

2.1 PYROLYSIS OF PETROLEUM RECOVERY FROM VARIOUS PETROLEUM PROCESSES

In this process styrene is separated from thermally cracked petroleum. In the process of this invention thermally cracked petroleum is initially distilled to recover the fraction boiling between 120°C and 160°C. This fraction is then subjected to extractive distillation with an organic polar solvent containing a nitrile polymerization inhibitor in which the styrene is soluble. The solvent is removed and the styrene containing fraction is thereafter treated with nitric acid after which it is scrubbed with water or alkali. The resulting product is fractionated to recover styrene of high purity which is substantially colourless.

- This process is not favourable due to the unavailability of raw material
- Due to carbon in petroleum, catalyst can also be poisoned.

2.2 SIDE CHAIN CHLORINATION OF ETHYLBENZENE HYDROLYSIS TO THE CORRESPONDING ALCOHOLS FOLLOWED BY DEHYDRATION AND SIDE CHAIN CHLORINATION OF ETHYLBENZENE FOLLOWED BY DECHLORINATION

The process involves chlorination of ethylbenzene in the presence of heat to form partial intermediate i.e. ethylbenzene chloride which is then treated with water to form ethyl benzaldehyde.

This process are not used generally because

- It suffers from high cost of raw materials.
- It forms the chlorinated contaminants in the monomer.

2.3 CATALYTIC DEHYDROGENATION OF ETHYLBENZENE

The majority of industrial production of styrene follows from the dehydrogenation of ethyl benzene. This dehydrogenation process involves the catalytic reaction of ethylbenzene.

REACTION



PROCESS DESCRIPTION

The fresh ethyl benzene is mixed with the recycled ethylbenzene then it is mixed with steam and is fed to the primary and secondary dehydrogenation reactors. When steam exits the re-heater as a cool product, it is then reheated to superheated steam which then enters the primary dehydrogenation reactor. Superheated steam is used for heating the mixture for the reaction in the secondary dehydrogenation reactor as well. These dehydrogenation reactors are designed to have low pressure and an even flow distribution. The dehydrogenation reactors liquid waste is then cooled through a number of heat exchangers which heat the recycled and fresh ethylbenzene. The reactors liquid waste is divided and condensed which then enters a vent gas compressor which further minimized pressure drop. This process has three distillation towers that operate to reduce polymer formation, to have low temperatures and to operate under vacuum conditions.

- This process is favourable because the process reaction is equilibrium limited.
- With the addition of steam, the process can be controlled and steam is recycled. So, this process is economical.

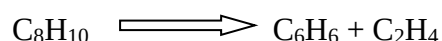
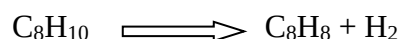
This process is primary commercialize process for production of styrene about 85% of the industrial process used nowadays. Ethyl benzene is reacted with catalyst usually iron oxide. During the process, the steam does not react with ethyl benzene and the catalyst which prevent coking from happen. The advantage of diluting ethyl benzene with superheated steam in this process are:

- It lowers the partial pressure of ethyl benzene and shift of equilibrium towards higher styrene production and minimizing the loss to thermal cracking.
- Decrease carbonaceous deposits by steam reforming reaction.

CHAPTER 3

MATERIAL BALANCE

3.1 REACTION



3.2 SPECIFICATIONS

Conversion of ethylbenzene	
Conversion of ethylbenzene in reactor 1	30%
Conversion of ethylbenzene in reactor 2	30%
Overall conversion	60%
Selectivity	
Styrene, Hydrogen	95%
Benzene, Ethylene	3%
Toluene, Methane	2%
Molar Ratio	
Superheated steam/ ethylbenzene in reactor 1	6:1
Superheated steam / ethylbenzene in reactor2	8:1

3.3 MATERIAL BALANCE CALCULATION

BASIS: 250 TON / DAY

Styrene in product = $250 \times 1000 / 24 \times 104.15 = 100 \text{ Kmol / hr}$

As our conversion is 60%

Ethylbenzene required = $100 / 0.6 = 166.66 \text{ Kmol / hr}$

The selectivity of the reaction is 95%

Hence, ethylbenzene required = $166.66 / 0.95 = 175.4 \text{ Kmol / hr}$

Efficiency of separator = 98%

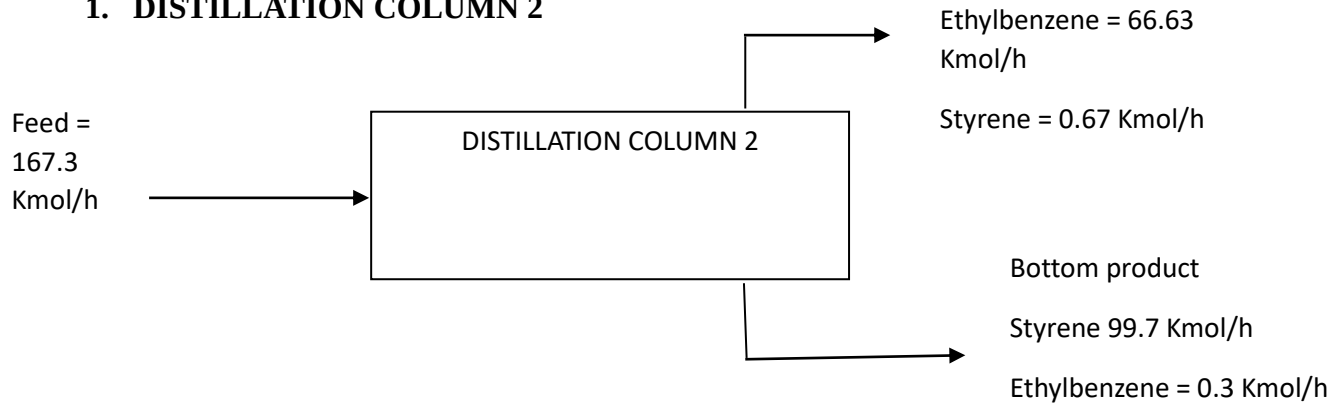
Efficiency of the styrene column = 99.7%

Therefore, ethylbenzene required = $175.4 / (0.98 \times 0.997 \times 0.95) = 180 \text{ Kmol / hr}$

- Selectivity for production of toluene, methane as side reaction is 2%
 The produced moles of toluene = $180 \times 0.6 \times 0.02 = 2.16 \text{ Kmol / hr}$
 The produced moles of methane = 2.16 Kmol / hr

- Selectivity of production of benzene, ethylene as side reaction is 3%
The produced mole of benzene = $180 * 0.6 * 0.03 = 3.24 \text{ Kmol / hr}$
The produced moles of ethylene = 3.24 Kmol / hr

1. DISTILLATION COLUMN 2



$$\text{Overall balance } A = B + C \quad (1)$$

$$\text{Styrene balance: } w \cdot A = w \cdot B + w \cdot C \quad (2)$$

$$\text{Ethylbenzene balance: } x \cdot A = x \cdot B + x \cdot C$$

$$\text{Styrene in product} = 100 \text{ Kmol / hr}$$

Purity of styrene at bottom is 99.7% and ethylbenzene at the top of column is 99% pure.

Our conversion is 60%

This implies, that feed stream the styrene is 60% and remaining 40% is ethylbenzene.

$$A = B + C$$

$$0.60 \cdot A = 0.01 \cdot B + 0.997 \cdot 100$$

$$0.60 \cdot A = 0.01 \cdot B + 99.7$$

Using (1) in (2), we get

$$0.60 \cdot (B + 100) = 0.01 \cdot B + 99.7$$

$$0.59 \cdot B = 39.7$$

$$B = 67.3 \text{ Kmol / hr}$$

$$\text{Now, } A = 67.3 + 100 = 167.3 \text{ Kmol / hr}$$

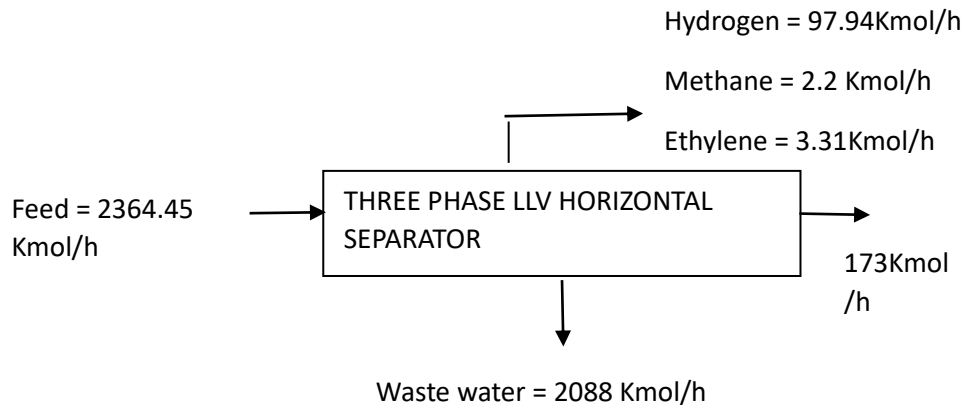
2. DISTILLATION COLUMN 1

At the inlet of the distillation column

- The produced moles of toluene = $180 * 0.6 * 0.02 = 2.16 \text{ Kmol/ hr}$
- The produced moles of benzene = $180 * 0.6 * 0.03 = 3.24 \text{ Kmol/ hr}$
- The top product (B) = $3.24 + 2.16 = 5.4 \text{ Kmol/hr}$
- Hence, the inlet of distillation column 1 = $167.3 + 5.4 = 173 \text{ Kmol/h}$

3. SEPARATOR

u= waste water
 v= gases
 w= styrene
 x= ethylbenzene
 y= toluene
 z= benzene



steam added to first reactor = moles of ethylbenzene * 6 = 1080 Kmol/h.

As 30% conversion in 1st reactor

Steam added to the second reactor = moles of ethylbenzene * 8 = 1008 Kmol/h

Total steam added = 1080 + 1008 = 2088 Kmol/h

Styrene at the inlet of the separator = $100 / (0.98 * 0.997) = 102.34 \text{ Kmol/h}$

Ethylbenzene at the inlet of the separator = $67.3 / (0.98 * 0.997) = 69 \text{ Kmol/h}$

Toluene at the inlet of the separator = $2.16 / 0.98 = 2.2 \text{ Kmol/h}$

Methane at the inlet of the separator = 2.2 Kmol/h

Hydrogen at the inlet of the separator = $102.34 - 2 * (2.2) = 97.94 \text{ Kmol/h}$

Benzene at the inlet of the separator = 3.31 Kmol/h

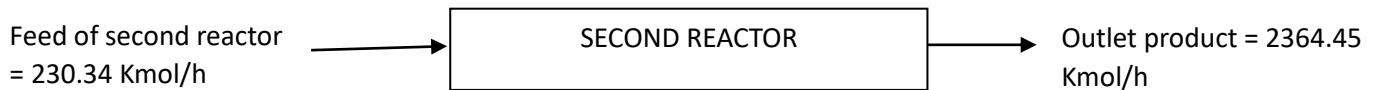
Ethylene at the inlet of the separator = 3.31 Kmol/h

Hence, the total feed of separator = A= 2364.45 Kmol/h (including steam)

4. SECOND REACTOR

CONVERSION OF 2ND REACTOR = 30%

As the steam is inert in reaction so ignoring it in reactor balance



Total amount of outlet of second reactor = B = 2364.45 Kmol/h

Outlet of second reactor

Styrene	102.34 Kmol/h
Ethylbenzene	69 Kmol/h
Toluene	2.2 Kmol/h
Methane	2.2 Kmol/h
Benzene	3.31 Kmol/h
Ethylene	3.31 Kmol/h
Hydrogen	97.94 Kmol/h

Inlet of second reactor

Styrene at the inlet of second reactor = $102.34 - (180 \times 0.30 \times 0.95) = 51.18$ Kmol/h

Ethylbenzene at the inlet of second reactor = 124.5 Kmol/h

Toluene at the inlet of second reactor = $2.2 \times 0.30 / (0.60 \times 0.95) = 1.15$ Kmol/h

Methane at the inlet of the second reactor = 1.15 Kmol/h

Benzene at the inlet of second reactor = $3.31 \times 0.30 / (0.60 \times 0.95) = 1.74$ Kmol/h

Ethylene at the inlet of the second reactor = 1.74 Kmol/h

Hydrogen at inlet of second reactor = $51.18 - 2 \times (1.15) = 48.88$ Kmol/h

Hence, the total inlet of second reactor = 230.34 Kmol/h

5. FIRST REACTOR

Conversion of first reactor = 30%

Selectivity is 95% for styrene reaction

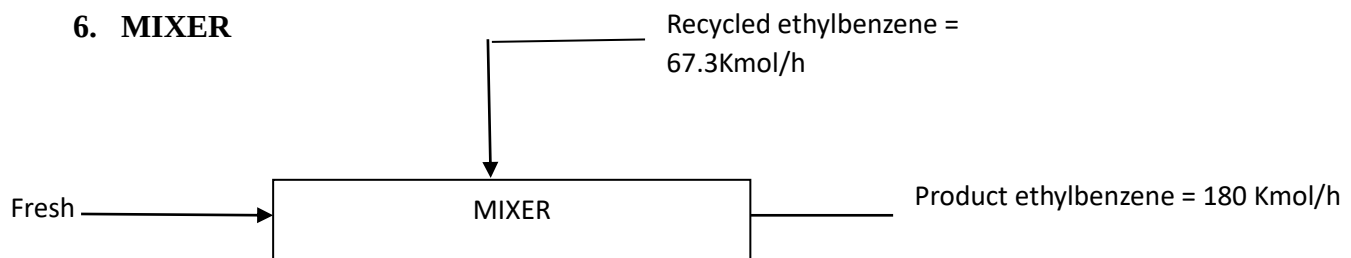
Outlet of first reactor

Styrene	51.15 Kmol/h
Ethylbenzene	124.5 Kmol/h
Toluene	1.15 Kmol/h
Methane	1.15 Kmol/h
Benzene	1.74 Kmol/h
Ethylene	1.74 Kmol/h
Hydrogen	48.88 Kmol/h

Inlet of first reactor

- Styrene at the inlet of the first reactor = $51.18 - (180 \cdot 0.30 \cdot 0.95) = 0.03$ Kmol/h
- Ethylbenzene at the inlet of first reactor
 Unreacted ethylbenzene = 124.5 Kmol/h
 Ethylbenzene at the inlet = $124.5 + (111 \cdot 0.30 / 0.6) = 180$ Kmol/h

6. MIXER



The fresh ethylbenzene = $180 - 67.3 = 112.7$ Kmol/h

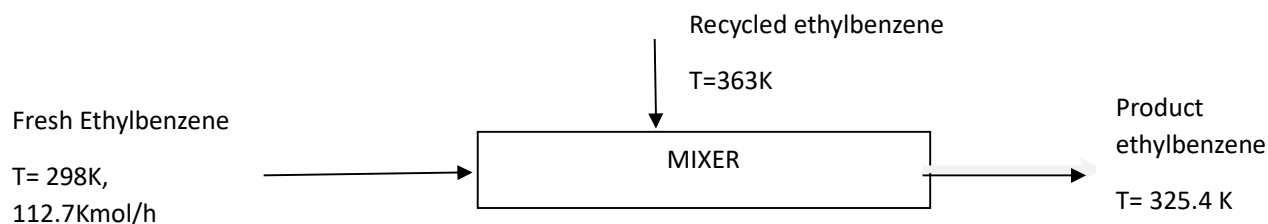
CHAPTER 4

ENERGY BALANCE

SPECIFIC HEAT TABLE ($C_p = A + BT + CT^2 + DT^3$)

COMPONENT	A	B	C	D
ETHYLBENZENE	-43.099	0.70715	-0.0004811	1.3008×10^{-7}
STYRENE	-28.248	0.61588	-0.0004023	9.9353×10^{-8}
BENZENE	-33.917	0.47436	-0.0003017	7.130×10^{-8}
TOLUENE	-24.355	0.51246	-0.0002765	4.9111×10^{-8}
ETHYLENE	3.806	0.15659	-0.00008348	1.7551×10^{-8}
METHANE	19.251	0.052126	0.000011974	-1.132×10^{-8}
HYDROGEN	27.143	0.009274	-0.00001381	7.6451×10^{-9}
WATER	32.243	0.001924	0.00001055	-3.596×10^{-9}

1. ENERGY BALANCE ON MIXER



Inlet of mixer

Compound	Molar flow rate (Kmol/h)	Temperature (K)	Specific heat capacity (KJ/Kmol*K)
Fresh ethylbenzene	112.7	298	128.35
Recycled ethylbenzene	67.3	363	156.42

Balance on Ethylbenzene

$$Q_{\text{recycled}} = Q_{\text{fresh}}$$

$$m_1 C_p \Delta T = m_2 C_p \Delta T$$

$$(67.3) \cdot (156.42) \cdot (363 - T) = 112.7 \cdot 128.35 \cdot (T - 298)$$

$$T = 325.4 \text{ K}$$

Hence, $Q = 112.7 \times 128.35 \times (325.4 - 298)$

$Q = 396342.23 \text{ KJ/h}$

2. ENERGY BALANCE ON HEAT EXCHANGER 1



Inlet of heat exchanger 1

Compound	Molar flow rate (Kmol/h)	Temp (K)
ethylbenzene	180	325.4

Outlet of heat exchanger 1

Compound	Molar flow rate (Kmol/h)	Temp (K)
ethylbenzene	180	400

$T(\text{mean}) = 325.4 + 400 / 2 = 362.7 \sim 363\text{K}$

Heat capacity at 363K = $\int C_p dT = \int A + BT + CT^2 + DT^3 = 156.53 \text{ KJ/Kmol} \cdot \text{K}$

$Q = mC_p \Delta T = 180 \times 156.53 \times (400 - 325.4) = 2101884.8 \text{ KJ/h}$

3. ENERGY BALANCE ON HEAT EXCHANGER 2



$T(\text{mean}) = 400 + 550 / 2 = 475\text{K}$

Heat capacity at 475K = $\int C_p dT = \int A + BT + CT^2 + DT^3 = 198.19 \text{ KJ/Kmol} \cdot \text{K}$

$Q = mC_p \Delta T = 180 \times 198.19 \times (550 - 400) = 5351130 \text{ KJ/h}$

4. ENERGY BALANCE ON HEAT EXCHANGER 3

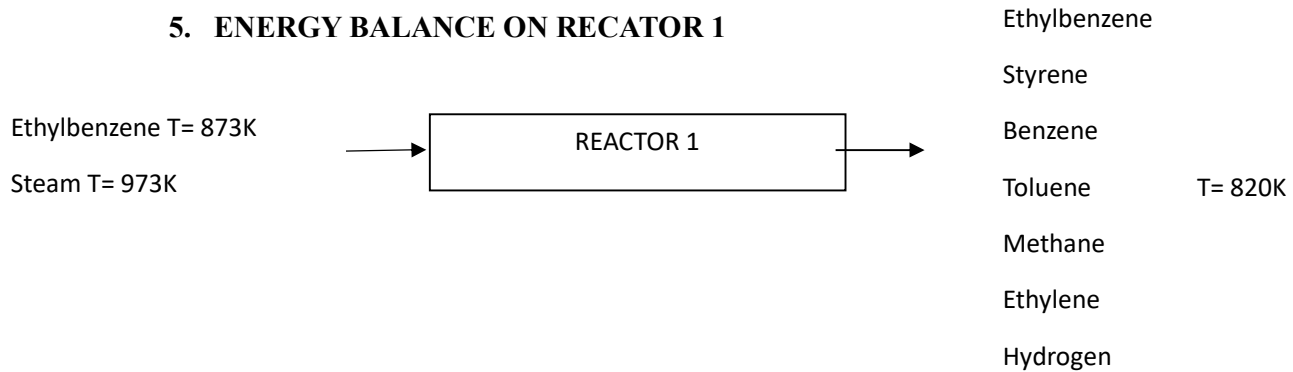


$$T(\text{mean}) = 550 + 873 / 2 = 711.5\text{K}$$

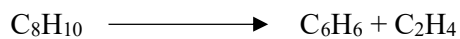
$$\text{Heat capacity at } 475\text{K} = \int C_p dT = \int A + BT + CT^2 + DT^3 = 198.19 \text{ KJ/Kmol} \cdot \text{K}$$

$$Q = mC_p\Delta T = 180 \cdot 711.5 \cdot (873 - 550) = 15310587 \text{ KJ/h}$$

5. ENERGY BALANCE ON REACTOR 1



REACTION



$$\Delta H_f = 117.53 * \frac{10^3 \text{KJ}}{\text{Kmol}}$$

$$\Delta H_f = 105.256 * \frac{10^3 \text{KJ}}{\text{Kmol}}$$

$$\Delta H_f = -54.694 * \frac{10^3 \text{KJ}}{\text{Kmol}}$$

For Outlet stream

$$\text{Heat capacity at } 873\text{K} = \int C_p dT = \int A + BT + CT^2 + DT^3$$

$$\Delta H_{\text{rex}} = (117.53 * 10^3 * 51.18) + (105.256 * 10^3 * 1.74) + (-54.694 * 10^3 * 1.15) \\ = \frac{6135438.74 \text{ KJ}}{h}$$

Inlet of reactor 1

Compound	Molar flow rate	Temperature (k)	Heat capacity Cp
Ethylbenzene	180	873	294.12
steam	1080	973	40.57

Outlet of reactor 1

compound	Molar flow rate	Cp (KJ/Kmol*K)	Mole fraction
ethylbenzene	124.5	294.123	0.0449
styrene	51.18	268.914	0.01883
Benzene	1.74	197.7	0.000851
Methane	1.74	83.564	0.00276
Ethylene	1.15	66.351	0.0024
Toluene	1.15	244.917	0.00048
Hydrogen	48.88	29.801	0.929
Steam	1080	40.57	

$$C_p(\text{mix}) = (294.123 * 0.0449) + (268.914 * 0.01883) + (197.7 * 0.000851) + (83.564 * 0.00276) + (66.351 * 0.0024) + (244.917 * 0.00048) + (29.801 * 0.929)$$

$$C_p(\text{mix}) = 46.62 \text{ KJ/Kmol*K}$$

Now,

$$Q_{\text{STEAM}} - \Delta H_f = m * C_p * \Delta T$$

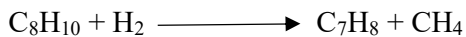
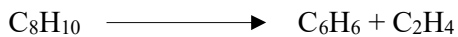
$$1080 * 40.57 * (973 - T) - 6135432.74 = 230.34 * 46.62 * (873 - T)$$

$$T = 820 \text{ K}$$

$$\text{And, } Q = 230.34 * 46.62 * (873 - 820) = 569137.892 \text{ KJ/h}$$

6. ENERGY BALANCE ON REACTOR 2

REACTION



For Outlet stream

$$\text{Heat capacity at } 820\text{K} = \int C_p dT = \int A + BT + CT^2 + DT^3$$

$$\Delta H_{\text{rex}} = (117.53 * 10^3 * 102.34) + (105.256 * 10^3 * 3.31) + (-54.694 * 10^3 * 2.2) \\ = \frac{12256090.8 \text{ KJ}}{h}$$

Inlet of reactor 2

compound	Molar flow rate	Cp (KJ/Kmol*K)	Mole fraction
ethylbenzene	124.5	286.14	0.0136
styrene	51.18	261.15	0.00571
Benzene	1.74	192.49	0.0002581
Methane	1.74	63.80	0.000837
Ethylene	1.15	85.76	0.000721
Toluene	1.15	237.01	0.000145
Hydrogen	48.88	29.67	0.282
Steam	1080	38.94	0.696
Fresh steam	1088 (973K)	40.80	

$$C_p(\text{mix}) = (286.14 * 0.0136) + (261.15 * 0.00571) + (192.49 * 0.0002581) + (63.80 * 0.000837) + (85.761 * 0.000721) + (237.01 * 0.000145) + (29.67 * 0.282) + (38.94 * 0.696)$$

$$C_p(\text{mix}) = 41.04 \text{ KJ/Kmol*K}$$

Now,

$$Q_{\text{STEAM}} - \Delta H_f = m * C_p * \Delta T$$

$$1088 * 40.80 * (973 - T) - 12256091 = 2448.3 * 41.04 * (T - 820)$$

$$T = 782\text{K}$$

$$\text{And, } Q = 2448.3 * 41.04 * (820 - 782) = 3818172.82 \text{ KJ/h}$$

7. ENERGY BALANCE ON HEAT EXCHANGER 4



$$T_{\text{mean}} = 782 + 500 / 2 = 641 \text{ K}$$

$$\text{Heat capacity at 641 K} = \int C_p dT = \int A + BT + CT^2 + DT^3$$

Compound	Molar flow rate	C _p (KJ/Kmol*K)	Mole fraction
Ethylbenzene	69	246.77	0.0292
Styrene	102.34	227.4	0.0433
Benzene	3.31	164.96	0.0014
Toluene	2.2	203.46	0.00093
Ethylene	3.31	74.50	0.0014
Methane	2.2	54.6	0.00093
Hydrogen	97.94	29.43	0.0414
Steam	2088	36.86	0.883

$$C_p(\text{mix}) = (246 \cdot 0.0292) + (227.4 \cdot 0.0433) + (164.96 \cdot 0.0014) + (203.46 \cdot 0.00093) + (74.5 \cdot 0.0014) + (54.6 \cdot 0.00093) + (29.43 \cdot 0.0414) + (36.86 \cdot 0.883)$$

$$C_p = 51.4 \text{ KJ/Kmol} \cdot \text{K}$$

$$Q = m \cdot C_p \cdot \Delta T$$

$$Q = 2364 \cdot 51.4 \cdot 282$$

$$Q = 34265707.2 \text{ KJ/h}$$

8. ENERGY BALANCE ON HEAT EXCHANGER 5



$$T_{\text{mean}} = 500 + 280 / 2 = 390 \text{ K}$$

Compound	Molar flow rate	C _p (KJ/Kmol*K)	Mole fraction
Ethylbenzene	69	167.23	0.0292
Styrene	102.34	156.69	0.0433
Benzene	3.31	109.43	0.0014
Toluene	2.2	136.38	0.00093
Ethylene	3.31	53.22	0.0014
Methane	2.2	40.73	0.00093
Hydrogen	97.94	29.11	0.0414
Steam	2088	34.38	0.883

$$\text{Heat capacity at 390 K} = \int C_p dT = \int A + BT + CT^2 + DT^3$$

$$C_p(\text{mix}) = (167.23 \times 0.0292) + (156.69 \times 0.0433) + (109.43 \times 0.0014) + (136.38 \times 0.00093) + (53.22 \times 0.0014) + (40.73 \times 0.00093) + (29.11 \times 0.0414) + (34.38 \times 0.883)$$

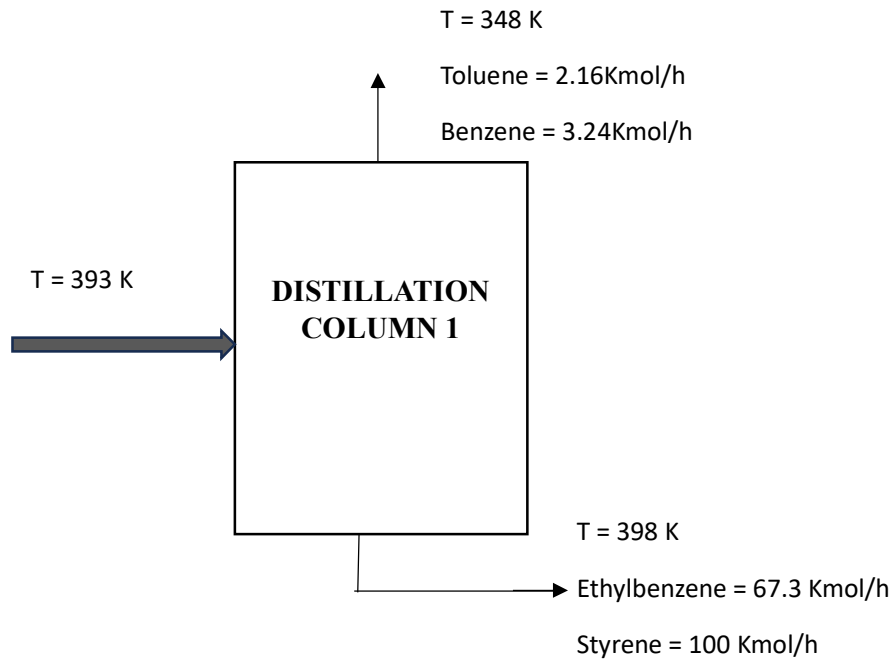
$$C_p = 43.6 \text{ KJ/Kmol} \cdot \text{K}$$

$$Q = m \cdot C_p \cdot \Delta T$$

$$Q = 2364 \times 43.6 \times 220$$

$$Q = 22675488 \text{ KJ/h}$$

9. ENERGY BALANCE ON DISTILLATION COLUMN 1



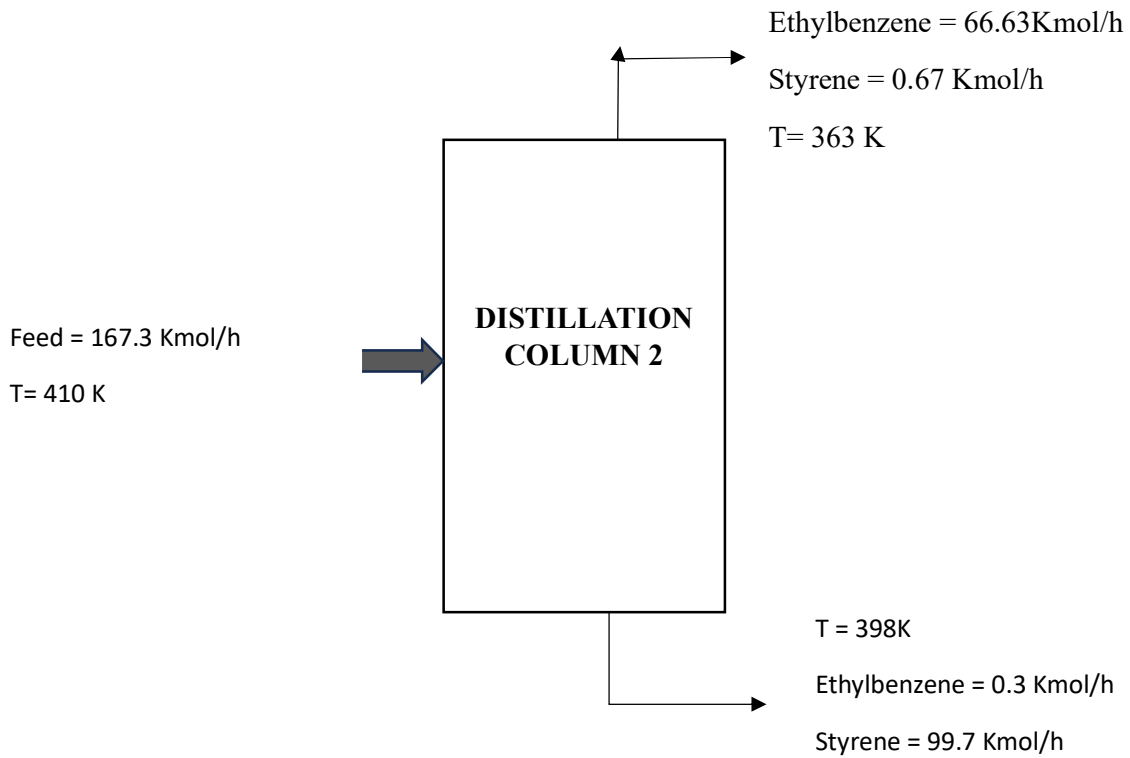
Outlet of distillation column 1

Component	MASS FLOW RATE (Kmol/hr)	C_p (KJ/Kmol K)	λ (KJ/Kmol)
C_8H_{10}	67.3	159.475	
C_6H_6	3.24	104.04	30781
C_8H_8	100.34	149.76	
C_7H_8	2.16	130.05	33201

$$Q = 67.3 \times 159.475 (398 - 393) + 100.34 \times 149.76 (398 - 393) + 3.24 \times 104.04 (348 - 393) + 2.16 \times 130.05 (348 - 393) - 2.16 \times 33201 - 3.24 \times 30781$$

$$Q = -67544.3 \text{ KJ/hr}$$

10. ENERGY BALANCE ON DISTILLATION COLUMN 2



Inlet of Distillation column 2

COMPONENT	MASS FLOW RATE (Kmol/hr)	C _p (KJ/Kmol K)	λ (KJ/Kmol)
C ₈ H ₁₀	67.3	175.092	35588
C ₈ H ₈	100	163.616	36844

$$Q_{\text{RELEASE}} = 99.7 \times 163.616 \times (398 - 410.1) + 67.3 \times 163.616 \times (363 - 410.1) + 66.63 \times 175.092 \times (363 - 410.1) + 0.3 \times 175.092 \times (398 - 410.1) - 67.63 \times 35588$$

$$Q_{\text{RELEASE}} = -24955061.5 \text{ KJ/hr}$$

CHAPTER 5

PROCESS EQUIPMENT DESIGN

MINOR EQUIPMENT DESIGN

HEAT EXCHANGER

5.1 DESIGN PROCEDURE

GIVEN DATA

SHELL SIDE	TUBE SIDE
Hot fluid	Cold fluid
Dowtherm a	Ethylbenzene
$T_1 = 180^\circ\text{C}$	$t_1 = 52^\circ\text{C}$
$T_2 = 130^\circ\text{C}$	$t_2 = 127^\circ\text{C}$
Mass flow rate = 21580 kg/h	Mass flow rate = 19109.7 kg/h

Step1. Obtain the required thermophysical properties of hot and cold fluids at the caloric temperature or arithmetic mean temperature. Calculate these properties at the caloric temperature if the variation of viscosity with temperature is large.

	Shell side	Tube side
Density Kg/m^3	947.8	803.64
Viscosity Pascal second	0.00056	0.000334
Specific heat capacity $\text{KJ/Kg}^\circ\text{K}$	1.954	1.837
Thermal conductivity $\text{W/m}^\circ\text{K}$	0.1171	0.1147

Step2: From Energy balance the heat duty of an exchanger 1

$$Q = 586432.65\text{W}$$

Step3: Assume a reasonable value of overall heat transfer coefficient ($U_{O \text{ assumed}}$). The value of ($U_{O \text{ ass}}$) With respect to the process hot and cold fluids can be taken from the books

From literature, $U_0(\text{assumed}) = 300 \text{ W/m}^2\text{K}$

Step 4: Determine the LMTD and the correction factor F_T . F_T normally should be greater than 0.75 For the steady operation of the exchangers. Otherwise it is required to increase the no of passes to obtain higher F_T value.

Assuming counter flow arrangement

$$LMTD = (T_1 - t_2) - (T_2 - t_1) / \ln(T_1 - t_2 / T_2 - t_1)$$

$$LMTD = 64.3^\circ\text{C}$$

$$R = (T_1 - T_2) / (t_2 - t_1) = 0.666$$

$$S = (t_2 - t_1) / (T_1 - t_1) = 0.585$$

From graph, $F_t = 0.85$

$$(LMTD)_{TRUE} = 0.85 * 64.3^\circ\text{C} = 54.6^\circ\text{C}$$

Step5: Calculate Heat transfer Area

$$Q = UA(LMTD)_{true}$$

$$A = Q / U * (LMTD)_{true}$$

$$A = 586432.65 / 300 * 54.6 = 36\text{m}^2$$

Step 6: Select the tube material, decide the tube diameter (ID= d_i , OD= d_o), its wall thickness (in terms of BWG or SWG) and tube length (L). Calculate the numbers of tubes (N_T) required to provide heat transfer area (A).

$$N_T = A / \pi * d_o * L$$

$$N_T = 164 \text{ tubes}$$

Hence, number of tubes per pass = 41

Step 7: Calculate Tube side velocity

- Side cross sectional area = $\pi/4 * d^2 = 0.0001778 \text{ m}^2$
- Area per pass = $0.0001778 * 41 = 0.007289 \text{ m}^2$
- Tube side velocity = mass flow rate / density * area per pass = 0.91 m/s

Tube outer diameter = 19.05mm
Length = 12ft
Tube thickness = 2mm
1-4 shell and tube heat exchanger

Step 8: Decide type of shell and tube heat exchanger (fixed tube sheet, u tube, floating head exchanger). Select the tube pitch (P_T), determine inside shell diameter (D_e) that can accommodate the number of tubes (N_T).

- Since the fouling resistance = $0.0002 \text{ m}^2 \text{ C h} / \text{W}$. Therefore, the floating head type heat exchanger is employed.
- Triangular pitch is used.

Determination of inside Shell Diameter (D_e)

Shell diameter (D_s) = bundle diameter + shell bundle clearance

- Bundle diameter = $d_0 \cdot (N_T/K_1)^{1/n_1} = 0.3824\text{m}$
Using graph between bundle diameter and clearance, the shell diameter (D_s) = 0.437m

Step 9: Determine the Tube side Heat transfer Coefficient (h_i)

- Reynold number = (density*velocity*inner diameter)/ viscosity
Reynold no = 32952.8
- Prandtl number = (viscosity*specific heat capacity)/thermal conductivity
Prandtl number = 5.35
- Nusselt number = heat transfer factor * Reynold no *(Prandtl no) ^0.33
Nusselt number = $3.5 \cdot 10^{-3} \cdot 32952.8 \cdot (5.35)^{0.33} = 200.59$
- Tube side heat transfer coefficient = Nusselt no * thermal conductivity/ inner diameter

Tube side heat transfer coefficient = 1528.77 W/m²K

Step10: Determine Tube side Pressure Drop

$$\Delta P = Np \left(8 * jf * \left(\frac{l}{di} \right) + 2.5 \right) * \rho v^2 / 2$$

$$\Delta P = 4 \left(8 * 3.8 * 10^{-3} * \left(\frac{3.66}{0.01505} \right) + 2.5 \right) * 803.64 * \frac{(0.91)^2}{2}$$

$$\Delta P = 0.13 \text{ bar}$$

Step 11: shell side calculation

- Baffle spacing (l_b) = 0.5* shell diameter = 0.5* 0.437 = 0.2185m
- $P_t = 1.25 * d_0 = 1.25 * 0.01905 = 0.0238125\text{m}$
- Cross flow area $A_s = \frac{(P_t - d_o) * D_s * l_b}{P_t} = 0.0191\text{m}^2$
- Shell side velocity = mass flow rate/ density *cross flow area
= 21580/947.8*0.0191
= 0.33m/s
- Shell side equivalent diameter $d_e = \frac{1.10}{d_o} * (P_t^2 - 0.917d_o^2)$
= 0.013486m
- Reynold number = $\rho v d_e / \mu = 947.8 * 0.33 * 0.013486 / 0.00056$
= 7562.4
- Prandtl number = $\mu C_p / k = 0.00056 * 1.954 / 0.1171 = 9.34$
- Nusselt number = $j_h * \text{Reynold number} * (\text{Prandtl number})^{0.33}$
Nusselt number = $5.5 * 10^{-3} * (7562.4) * (9.34)^{0.33}$
= 86.943
- **Shell Side Heat transfer coefficient = $Nu * k / d_e$**
= **$86.943 * 0.1171 / 0.01348$**
= **751.92**

- Overall heat transfer coefficient (U_o calculated) based on outside tube area including dirt factors.

Fouling factor of Dowtherm (R_{do}) = 0.0002 m²C/W

Fouling factor of fatty acid (R_{di}) = 0.0002 m²C/ W

$$U_{o,cal} = \left[\frac{1}{h_o} + R_{do} + \frac{A_o}{A_i} \left(\frac{d_o - d_i}{2k_w} \right) + \frac{A_o}{A_i} \left(\frac{1}{h_i} \right) + \frac{A_o}{A_i} R_{di} \right]^{-1}$$

$$U_{o,calculated} = 376.22 \text{ W/m}^2\text{C}$$

- % overdiseign = IUO calculated – UO assumed / UO assumed I x 100
- % overdiseign = (376.22 – 300) / 300
- % overdiseign = 25.4% <30%

Therefore, the calculated overall heat transfer coefficient is well within the design criteria

- **Shell Side Pressure Drop**

$$\Delta P = 8 * Jf * \left(\frac{Ds}{de} \right) * \left(\frac{L}{Lb} \right) * \rho * v^2 / 2$$

$$\Delta P = 0.11 \text{ bar}$$

5.2 SPECIFICATION SHEET

SHELL AND TUBE HEAT EXCHANGER

Section	Parameter	Value	Unit
1. Fluid Allocation	Tube Side (Cold fluid)	Ethylbenzene	-
	Shell Side (Hot fluid)	Dowtherm a	-
	Mass flow rate (Tube side)	19109.7	Kg/hr / Kg/s
	Mass Flow rate (Shell side)	21580	Kg/hr / Kg/s
	Specific heat (Tube/shell)	1.837 / 1.954	KJ/Kg K
	Inlet Temp (Tube/shell)	52 / 180	°C
	Outlet Temp (Tube/shell)	127 / 130	°C
2. Heat Duty	Q	172560	W
3. Fluid Properties	Density (Tube/shell)	803.64/ 947.8	Kg/m ³
	Viscosity (Tube/shell)	0.000334 / 0.00056	Pa s
	Thermal Conductivity (Tube/shell)	0.1147 / 0.1171	W/m K
	Specific Heat (Tube/shell)	1837/ 1954	J/kg K
4. LMTD Calculation	ΔT LMTD	64.3	°C
	Correction Factor (Ft)	0.85	-
	Effective ΔT (ΔT true)	54.6	°C
5. Overall Heat Transfer Coefficient	Assumed U	300	W/m ² K
	Calculated U	376.22	W/m ² K
6. Heat Transfer area	Required Area	36	m ²
7. Tube Data	Tube Outer Diameter (do)	19.05	mm
	Tube inner Diameter (di)	15.05	mm
	Tube Length	3.66	m
	Number of Tubes	164	-

	Layout	4-pass (41 tubes/pass)	-
	Tube Pitch	0.0238	m
	Baffle spacing	0.2185	m
8. Shell Design	Tube Bundle Diameter	382.4	mm
	Clearance	54.6	mm
	Shell Diameter	437	mm
9. Tube side Calculations	Flow Area Per pass	0.007289	m ²
	Velocity	0.91	m/s
	Reynolds number	32952.8	-
	Prandtl number	5.35	-
	Nusselt number	200.59	-
	Heat transfer coefficient (h _i)	1528.77	W/m ² K
10. Shell Side Calculations	Equivalent Diameter (d _e)	0.01354	
	Reynolds Number	7562.4	-
	Prandtl Number	9.34	-
	Nusselt Number	86.943	-
	Heat transfer coefficient (h _o)	751.92	W/m ² K
11. Pressure Drop	Tube side Pressure Drop (ΔP _t)	0.13	Bar
	Shell Side Pressure Drop(ΔP _s)	0.11	bar
12. MOC	Carbon steel		

MAJOR EQUIPMENT DESIGN

LLV SEPARATOR

5.3 DESIGN PROCEDURE

DATA GIVEN:

Liquids at outlet	Molar flow rate (Kmol/h)	Molar mass	Mole fraction	Mass flow rate (kg/h)
Styrene	102.34	104.15	0.045	10658.7
Ethylbenzene	69	106.167	0.030	7325.52
Toluene	2.21	92.14	0.00097	203.63
Benzene	3.31	78.11	0.00146	258.54
Waste water	2088	18.015	0.921	37615.32
Total	2264.86			56061.71

Gases at outlet	Molar flow rate (Kmol/h)	Molar mass	Mole fraction	Mass flow rate (kg/h)
Hydrogen	97.94	2.02	0.946	197.83
Methane	2.21	16.04	0.0213	34.45
Ethylene	3.31	28.05	0.032	92.85
Total	103.46			325.13

1. Calculation of liquid density

$$\rho_l = (909 * 0.045) + (866 * 0.030) + (867 * 0.00097) + (876 * 0.00146) + (1000 * 0.921)$$

$$\text{Liquid density} = 990 \text{ Kg/m}^3$$

2. Calculation of gas density

$$\rho_g = (0.082 * 0.946) + (0.656 * 0.0213) + (1.18 * 0.032)$$

$$\text{Hence, gas density} = 0.13 \text{ Kg/m}^3$$

As the vapors are less than 10 percent by weight so, horizontal separator will be recommended.

3. Calculation of settling velocity of the liquid droplets

$$U_t = k \sqrt{\frac{\rho_l - \rho_g}{\rho_g}}$$

$$U_t = 0.07 \sqrt{\frac{990 - 0.13}{0.13}}$$

Hence the settling velocity, $U_t = 6.12 \text{ m/s}$

Demister pads is used in this separator therefore, $U_a = U_t$

4. Vapor volumetric flow rate = Mass flow rate / (3600* gas density)
 $= 325.13 / (3600*0.13) = 0.696 \text{ m}^3/\text{sec}$

Taking, $h_v = 0.5 \times \text{vessel diameter}$

And as the operating pressure of LLV separator in the production of styrene is 10 bar. Hence, $L_v/D_v = 3$

5. Cross sectional area for vapor flow $= \frac{\pi}{4} * D_v^2 * 0.5$
 $= 0.393 D_v^2$
6. Vapor velocity = vapor volumetric flow rate/ cross sectional area for vapor flow
 $= 0.696 / 0.393 D_v^2$
 $= 1.771 / D_v^2$
7. Vapor residence time required for the droplet to settle to liquid surface $= h_v / U_a$
 $= 0.5 * D_v / 6.12$
 $= 0.082 * D_v$
8. Actual Residence time = vessel length / vapor velocity
 $= L_v / U_v = 3 * D_v / 1.771 * D_v^{-2}$
 $= 1.694 * D_v^3$
9. For satisfactory separation, required residence time = Actual residence time
 $0.082 * D_v = 1.694 D_v^3$
 $D_v = 0.22 \text{ m}$
10. Calculation of liquid volumetric flow rate $= 56061.71 / 3600 * 990$
 $= 0.0157 \text{ m}^3 / \text{sec}$
11. Liquid Cross Sectional area $= \frac{\pi}{4} * D_v^2 * 0.5$
 $= \frac{3.14}{4} * 0.22^2 * 0.5$
 $= 0.018 \text{ m}^2$
12. Length of vessel $= L_v = 3 * 0.22 = 0.66 \text{ m}$
13. Liquid Hold up volume $= 0.018 * 0.66 = 0.012 \text{ m}^3$
14. Liquid Hold up time = liquid volume / liquid flow rate
 $= 0.012 / 0.0157 = 0.78 \text{ sec}$

This is unsatisfactory very small hold up time and need to increase the liquid volume and this is best done by increasing the vessel diameter.

Now, $D_v = 1\text{ m}$

$$\begin{aligned} 15. \text{ Liquid Cross Sectional area} &= \pi / 4 * D_v^2 * 0.5 \\ &= \pi / 4 * 1^2 * 0.5 = 0.392\text{ m}^2 \end{aligned}$$

$$\text{Vessel length} = 1 * 3 = 3\text{ m}$$

$$16. \text{ Liquid Hold up volume} = 0.392 * 3 = 1.176 \text{ m}^3$$

$$17. \text{ Liquid Hold up time} = 1.176 / 0.0157 = 75 \text{ sec} = 1.25 \text{ min}$$

Not a satisfactory result

Therefore, taking $D_v = 1.4 \text{ m}$

$$\begin{aligned} 18. \text{ Liquid Cross Sectional area} &= \pi / 4 * D_v^2 * 0.5 \\ &= \frac{\pi}{4} * 1.4^2 * 0.5 \\ &= 0.769 \text{ m}^2 \end{aligned}$$

$$19. \text{ Vessel length} = 3 * 1.4 = 4.2 \text{ m}$$

$$20. \text{ Liquid Hold up volume} = 0.769 * 4.2 = 3.23 \text{ m}^3$$

$$\begin{aligned} 21. \text{ Liquid Hold up time} &= \text{hold up volume} / \text{liquid flow rate} \\ &= 3.23 / 0.0157 \\ &= 206 \text{ sec} = 3.43 \text{ min} \end{aligned}$$

Taking $D_v = 1.5 \text{ m}$

$$\begin{aligned} 22. \text{ Liquid Cross Sectional area} &= \pi / 4 * D_v^2 * 0.5 \\ &= \frac{\pi}{4} * 1.5^2 * 0.5 \\ &= 0.883 \text{ m}^2 \end{aligned}$$

$$23. \text{ Vessel length} = 3 * 1.5 = 4.5 \text{ m}$$

$$24. \text{ Liquid Hold up volume} = 0.883 * 4.5 = 3.974 \text{ m}^3$$

$$\begin{aligned} 25. \text{ Liquid Hold up time} &= \text{hold up volume} / \text{liquid flow rate} \\ &= 3.974 / 0.0157 \\ &= 253.12 \text{ sec} = 4.22 \text{ min} \end{aligned}$$

Hence, the new liquid residence time = 4.22 min. This is a satisfactory result as the residence time for a LLV separator used in the styrene production lie between 2-5 minutes.

5.4 SPECIFICATION SHEET

LLV HORIZONTAL SEPARATOR

Section	Parameter	value
Design Basis	Vessel type	Horizontal, three phase LLV
	Orientation	Horizontal
	Demister type	Mesh pad (standard)
	Number of separated phases	3(gas, organic liquid, waste water)
Liquid outlet	Styrene	102.34 Kmol/h
	Ethylbenzene	69 Kmol/h
	Toluene	2.21Kmol/h
	Benzene	3.31Kmol/h
	Waste water	2088 Kmol/h
	Total liquid	2264.86 Kmol/h
Gases outlet	Hydrogen	97.94 Kmol/h
	Methane	2.21 Kmol/h
	Ethylene	3.31 Kmol/h
	Total gases	103.46 Kmol/h
Physical and flow Properties	Liquid density	990 Kg/m ³
	Gas density	0.13 Kg/m ³
	Operating Pressure	10 bar
	Liquid volumetric flow rate	0.0157 m ³ /sec
	Vapor volumetric flow rate	0.696 m ³ /sec
	Settling velocity	6.12 m/s
	Liquid Hold up volume	3.974 m ³
	Liquid Hold up time	4.22 min
Vessel dimensions	Vessel Diameter	1.5m
	Liquid Height	0.75m
	Vessel length	4.5 m
	Length to diameter ratio	3
Design Features	Phase separation mechanism	Gravity + demister
	Retention time considered	yes

CHAPTER 6

COST ESTIMATION AND ECONOMICS

Acceptable plant design must present a process that is capable of operating under conditions, which will yield profit. Since net profit equals total value minus all expenses, it is essential that the chemical engineer be aware of the many different types of cost involved in the manufacturing processes. Capital must allocate for the direct, plant expenses, such as those for raw material, labour and equipment.

Besides direct expenses many others indirect expenses are incurred, and these must be included if a complete analysis of the total cost is to be obtained. Some examples of these indirect expenses are administrative salary, product distribution cost and cost for interplant communication. A capital investment is required for every industrial process and determination of necessary investment is an important part of a plant design process. The total investment for any process consist fixed capital investment for practical equipment and facilities in the plant plus working capital, which must be available to pay salaries, keep raw material and products on hand, and handle other special items requiring the direct cost outline.

When the cost for any type of commercial process is to be determined, sufficient accuracy has to be provided for reliable decision. There are many factors affecting investment and production cost. These are;

1. source of equipment
2. price fluctuation
3. company policies
4. operating and rate of production
5. governmental policies

Before an industrial plant can be put into operation, a large sum of money must be supplied to purchase and install the necessary machinery and equipment. Land and service facilities must be obtained, and the plant must be erected completely with all piping, controls and services.

The capital needed to supply the necessary manufacturing and plant facilities is called the fixed-capital investment, while that necessary for the operation of plant is termed the working capital. The sum of the fixed capital investment and the working is known as the total capital investment. Generally, the working capital amounts 10-20% of the total capital investment. Following is the breakdown of the fixed capital investment for a chemical process.

DIRECT COST:

1. purchased equipment
2. purchased equipment installation
3. instrumentation and control
4. piping
5. electrical equipment and material
6. building (including services)
7. yard improvement
8. land

INDIRECT COST:

1. engineering supervision
2. construction expenses
3. contractor's fee
4. contingency

TYPES OF CAPITAL COST ESTIMATE:

- Order of magnitude estimate (ratio estimate) based on similar cost data; probable accuracy of this estimate over $\pm 30\%$.
- Study estimate based on knowledge of major items of equipment, probable accuracy of this estimate up to $\pm 30\%$.
- Preliminary estimate (budget authorization estimate scope method): based on sufficient data to permit the estimate to the budget, probable accuracy of this estimate is within $\pm 20\%$.
- Detailed estimate based on complete engineering drawing, specifications and site survey, probable accuracy of this estimate within $\pm 10\%$.

5.1 ESTIMATION OF PRESENT PLANT COST

1. Fixed Capital Investment (FCI) Estimation Given:

Original plant capacity = 30000 TPA

Original capital cost (C original) = 100 million USD

Current plant capacity = 250 TPD = $250 \times 300 = 75000$ TPA

Cost index (Year: 2016) = 1582.3

Cost index at present (Year: 2025) = 2309.7

Formula Used:

Cost (2025) = Cost (2016) $\times$ (Capacity present / Capacity original) $^{0.6}$ $\times$ (Index present / Index original) Calculation:

Cost (2025) = $100 \times (75000/30000) ^{0.6} \times (2309.7/1582.3)$ Cost (2025) = 2153 crore.

Fixed capital investment required for 250 Tonne per day is 2153 crore.

DIRECT COST (70-85% of FCI)

They include the following

A. Equipment + Installation+ Instrumentation + Piping + Electrical + Insulation+
= 50% FCI

1) Purchased equipment (15-40% of FCI)

= 25% (FCI) = 538.25 crore

2) Installation including insulation and painting (25-55% of PEC)

= 30% (PEC) = 161.475 crore

3) Instrumentation and controls, installed (6-30% of PEC)

= 25% (PEC) = 134.56 crore

4) Piping installed (10-80% of PEC)

= 20% (PEC) = 107.65 crore

5) Electrical Installed (10-40% of PEC)

= 25% (PEC) = 134.56 crore

B. Buildings, Process and auxiliary (10-70% of PEC)

= 40% (Purchased equipment cost)

= 215.3 crore

C. Service facilities and yard improvement (40-100% of PEC)

= 60% (Purchased Equipment cost) = 323 crore

D. Land (4-8% Of PEC) = 5%(PEC) = 26.9125 crore

Total Direct cost = 1641.70 crore

INDIRECT COST (15-30% of FCI)

They include expenses which are not directly related with material or labour of actual installation of complete facility

A. Engineering and supervision (5-30% of Direct cost) = 10% (Direct cost) = 164.17 crore

B. Construction expenses and contractor's fee (6-30% of Direct cost) = 10% (Direct cost) = 164.17 crore

C. Contingency (5-15% of FCI) = 5%(FCI) = 107.65 crore

Total Indirect cost = Rs 435.99 crore

FIXED CAPITAL INVESTMENT = DIRECT COST + INDIRECT COST = 2077.69 crore

Working Capital: (10-20% of TCI) = 15% of FCI = 311.654 crore

Total Capital Investment = Fixed capital investment + Working Capital Investment
= 2389 crore

ESTIMATION OF TOTAL PRODUCT COST (TPC)

Manufacturing cost:

A. Fixed Charges (10-20% of TPC)

1. Depreciation (10% of FCI + 2-3% of building value for building) 10% of FCI + 2.5% of Building value = 213.152 crore

2. Local Taxes (1-4% of FCI)
4% of FCI = 83.11 crore

3. Insurance (0.4 – 1% of FCI)
0.7% of FCI = 14.54 crore

Total Fixed Charges = 310.802 crore

Total Product Cost (TPC) = Fixed Charges/0.15 = 2072.013 crore

B. DIRECT PRODUCTION COSTS

- Raw materials (10-50% of TPC) = 10% (TPC) = Rs. 207.2013 crore
- Operating labour (10-20% of TPC) = 10%(TPC)= Rs. 207.2013 crore
- Direct Supervisory and Clerical labour (10-25% of Operating labour) =

10% (Operating labour) = 20.72 crore

- Utilities (10-20% of TPC) = 10% (TPC) = Rs. 207.2013 crore
 - Maintenance and repairs (2-10% of FCI) = 3%(FCI) = Rs. 62.33 crore
 - Operating supplies (10-20% of cost for maintenance and repairs) = 15% (Maintenance and repairs) = Rs 15.58 crore
 - Laboratory charges (10-20% of Operating Labour) = 15% (Operating labour) = 31.1 crore
- Patents and royalties (0-6% of TPC) = 2%(TPC) = 41.44 crore

Total Direct Production cost = Rs. 793 Crore

C. Plant overhead Cost (5-10% of TPC) = 6% (TPC) = Rs. 124.3crore

Total Manufacturing Cost = 1228.102 crore

GENERAL EXPENSES:

Administration costs (2-6% of TPC) = 3% (TPC) = 62.16crore

Distribution and selling cost (2-20% of TPC) = 10% (TPC) = 207.2013 crore

Research and development = 5% (TPC) = 103.6 crore

Financing (0-10% of TCI) = 3% of TCI = 62.16 crore

Total General Expenses = 435.12 crore

Total Production cost = Total manufacturing cost + Total General Expenses= 1663.22 crore

SELLING PRICE

Styrene = 250/Kg

For a 330 days/year Plant

Total Income = Selling price × Quantity of product produced

= 2062.5 crore

Gross Earnings = Total income – Total Production cost

= Rs 399.28 crore

Let Taxes = 40% of gross earnings = 159.712 crore

Net Profit = Gross earning [1- tax rate] = Rs 239.6 crore

RATE OF RETURN = Net profit/ Total capital investment = $239.6/2389.34 = 10.03\%$

PAY BACK PERIOD =

Total capital investment/ net profit per year = $2389.34/ (239.6+213.152) = 5.3$ years

CHAPTER 7

SITE CONSIDERATION AND PLANT LAYOUT

The location of the plant can have a crucial effect on the profitability of a project, and the scope for future expansion. Factors considered while selecting a plant site are:

- Transportation
- Sources and costs of raw materials
- Prospective market for products
- Corporation long range planning
- Water source - quality and quantity
- Special incentive
- Climatic conditions
- Pollution requirements (Waste disposal)
- Utilities - cost, quantity and reliability; fuel - costs, reliability and availability
- Amount of site preparation necessary (site conditions)
- Construction costs
- Operating labour
- Taxes
- Living conditions
- Corrosion
- Expansion possibilities ☐
- Other factors.

Three factors are usually considered the most important. These are the location of the markets and raw materials and the type of transportation to be used.

1. **Transportation:** The transport of materials and products to and from the plant will be an overriding consideration in site selection. If practicable, a site should be selected that is close to at least two major forms of transport: road, rail, waterway (canal or river), or a sea port. The least expensive method of shipping is usually by water; the most expensive is by truck.
2. **Raw materials:** In the production of styrene, the availability of raw material is a primary factor influencing its site location. Styrene is Produced from ethylbenzene, which is synthesized using benzene and ethylene both are the derivative of petrochemical. Therefore, locating the plant near petrochemical complexes or refineries that produces these Feedstock ensures a stable and Cost effective supply.

3. **Location of markets:** Consumer products often are delivered in small shipment to a large number of customers. In an international market, there may be an advantage to be gained by locating the plant within an area with preferential tariff agreements. Since styrene is used in manufacturing of polystyrene, synthetic rubbers, and other raw material used in packaging, electronics, and construction, being near industrial clusters with plastic and polymer industries is advantageous.
4. **Long Range Corporate Planning:** The object of long range planning is to optimize a whole network of operations instead of each one individually. This means that each plant site is not considered only for itself and that its chosen location might not be the one that would be selected if only the economics of the one plant had been considered. Placing the plants throughout the country allows each plant to be located optimally.
5. **Climatic Conditions:** Adverse climatic conditions at a site will increase costs. Abnormally low temperatures will require the provision of additional insulation and special heating for equipment and pipe runs. Stronger structures will be needed at locations subject to high winds or earthquake.
6. **Pollution and Ecological Factors:** All industrial processes produce waste products, and full consideration must be given to the difficulties and cost of their disposal. The disposal of toxic and harmful effluents will be covered by local regulations, and the appropriate authorities must be consulted during the initial site survey to determine the standards must be met.
7. **Site Conditions:** An ideal chemical plant site is above the flood plain, flat, has good drainage, a high soil- bearing capability, and consists of sufficient land for the proposed plant and for future expansion.
8. **Availability of labour:** Access to a skilled and semi-skilled workforce is essential for operating a chemical plant. The location should be close to urban or semi-urban areas access to technical institutions, training centres, and housing for employees. Availability of labour also impact wage rates and employee retention.
9. **Political and Strategic considerations:** Capital grants, tax concession and other inducements are often given by governments to direct new investment to preferred locations; such as areas of high unemployment.
10. **Local Community Considerations:** The proposed plant must fit in with and be acceptable to the local community. On a new site, the local community must be able to provide adequate

facilities for the plant personnel: Schools, banks, housing and recreational and cultural facilities.

11. **Corrosion:** Once the general area for the plant has been determined, the effect of neighbouring industries should be considered when picking the specific site. Their presence may indicate an increased corrosion rate.

PLANT LAYOUT

The economic construction and efficient operation of a process depend on how well the plant and equipment specified on the process flow sheet is laid out. The principal factors to be considered are:

- Economic consideration: construction and operating costs.
- The process requirement.
- Convenience of operation.
- Convenience of maintenance.
- Safety.
- Future expansion.
- Modular construction.

Costs: The cost of construction can be minimized by adopting a layout that gives the shortest run of connecting pipe between equipment, and the least amount of structural steel works.

Operation: Equipment that needs to have frequent operator attention should be located convenient to the control room. Valves, sample points and instruments should be located at convenient positions and heights. Sufficient working space and headroom must be provided to allow easy access to equipment.

Safety: Blast walls may be needed to isolate potentially hazardous equipment, and confine the effects of an explosion. At least two escape routes for operators must be provided from each level in process buildings.

Plant expansion: Equipment placement should be Facilitate convenient tie-ins for future process expansion. Provision of ample space on pipe alloys and oversized services pipe accommodates future requirements.

Modular construction: In recent years there has been a move to assemble sections of plant at the plant manufacturer's site. These modules will include the equipment, structural steel, piping and instrumentation. The modules are then transported to the plant site, by road or sea. The advantages of modular construction are:

CHAPTER 8

INSTRUMENTATION AND PROCESS CONTROL

Instrumentation is the most important factor in ensuring safety and smooth working of the plant. A separate control room is provided in modern plants where on panels indicators and recorders are present. Instruments are used in the industry to measure process variables such as temperature, pressure density, level specific heat, conductivity, humidity, flow rate, chemical composition etc.

The primary objectives of the designer when specifying instrumentation and control schemes are:

1. Safe plant operation:
 - (a) To keep the process variables within known safe operating limits.
 - (b) To detect dangerous situations as they develop and provide alarms and automatic shutdown systems.
 - (c) To provide interlocks and alarms to prevent dangerous operating procedures.
2. Production rate: To achieve the design product output.
3. Product quality: To maintain the product composition within the specified quality standards.
4. Cost: To operate at the lowest production cost, commensurate with the other objectives.

These are not separate objectives and must be considered together.

TEMPERATURE CONTROL / MEASUREMENTS:

Various instruments are used for

- Thermocouple
- Resistance Thermometers
- Thermistors
- Mercury in Glass Thermometers.

Controllers are used to maintain temperature within specified limits. Every temperature control problems is essential one. The temperature lay involved in the measurement of

this Variable is an important factor. Thermal element is usually placed in a wall to protect it and allow servicing of element without interrupting the process. The location of the temperature element often has as much to do with the efficiency element often as other parts of the control loops. Temperature bulb should always be located at point where the coefficient of heat transfer will be as large as possible e.g. if vapours and liquid are at same temperature, then bulb should be kept in liquid because of high heat transfer coefficients.

PRESSURE CONTROL/ MEASUREMENTS:

It is quite necessary for most system handling vapours of gas. It can be measured by using Pressure gauges. Self operated pressure regulation is often used in pressure control. It is installed directly in the line the control sensing apparatus is placed about 10 diameter from the unit. This location eliminates erroneous pressure caused by turbulence. Sudden change in velocity, shock and vibrations difficulties often occur when self operated regulations are used with liquids.

LEVEL CONTROL/ MEASUREMENT:

The measurement of level can be defined as the determination of location of interface respect to a fixed plane. The main objective of a level control system is to maintain the liquid in a tank at the Act point value. The different is pressure transmitter senses pressure difference a function of liquid level in the tank) and gives out an electrical signal, after signal conditioning is given to the P.I.D. controller. The controller compares the measured variables with the set point and depending upon the error, gives an output to the control valve. The control value, in turn controls the flow. Due to the control of flow, the level is controlled to its set point.

FLOW CONTROL / MEASUREMENTS:

It is defined as volume per unit time at specific temperature and pressure conditions. It is generally measured by positive displacement of rate meters.

CONDENSER CONTROL:

Temperature control is unlikely to be effective for condensers, unless the liquid stream is sub-cooled. Pressure control is often used, or control can be based on the outlet coolant temperature.

DISTILLATION COLUMN CONTROL:

The primary objective of distillation column control is to maintain the specified composition of the top and bottom products, and any side streams; correcting for the effects of disturbances in:

- Feed flow-rate, composition and temperature.
- Steam supply pressure.
- Cooling water pressure and header temperature.
- Ambient conditions, which cause changes in internal reflux.

The compositions are controlled by regulating reflux flow and boil-up. The column overall material balance must also be controlled; distillation columns have little surge capacity (hold-up) and the flow of distillate and bottom product (and side-streams) must match the feed flows. The feed flow-rate is often set by the level controller on a preceding column. It can be Independently controlled if the column is fed from a storage or surge tank. Feed temperature is normally controlled, unless a feed pre heater is used. Temperature is often used as an indication of composition. The temperature sensor should be located at the position in the column where the rate of change of temperature with change in composition of the key component is a maximum.

Near the top and bottom of the column the change is usually small. With multi component systems, temperature is not a unique function of composition. Top temperatures are usually controlled by varying the reflux ratio, and bottom temperatures by varying the boil-up rate.

INSTRUCTIONS FOR OPERATION

Before starting, always make sure that the followings:

- No visible damage is evident in the system.
- All electrical switches are turned off.
- All water valves are closed.
- All valves are tightly closed.

Start- up procedure.

- Check the level in the tank.
- Switch on the main supply.
- Turn on the main pump.
- Open valve for pump.
- Start mechanical pump.
- Open the steam valve.
- Check the flow, temperature and pressure during the operation.

System shut down.

- Reduce the flow of the feed and steam.
- Reduce all flow rates and wait for pressure to drop.
- Turn off the pump.
- When the flow slow down close the valve tightly to avoid cavitations.
- Disconnect the packing line.

CHAPTER 9

POLLUTION CONTROL AND SAFETY

The Production of styrene involves several chemical processes, primarily the dehydrogenation of ethylbenzene, which requires high temperature and involves the handling of hazardous material. Due to the flammable, volatile and toxic nature of styrene and its intermediate, strict pollution control and safety measures are essential throughout the lifecycle of the product- from production and storage to handling and disposal.

1. NATURE AND HAZARDS OF STYRENE

Styrene is flammable, volatile organic compound with a sweet, aromatic odour. It poses health hazards such as irritation to the eyes, respiratory system, skin. Long term exposure can affect the nervous system.

2. AIR POLLUTION CONTROL

During styrene production, emission may arise from

- Volatile organic compounds from leaks and vents.
- Combustion gases from process heaters and furnace.

To control air pollution Vapour Recovery Units are used to collect and condense styrene vapours during loading and unloading. Activated Carbon Filters can be used in storage tank vents to capture residual vapours.

3. WASTE WATER TREATMENT

The production process generates waste water containing hydrocarbon, chemical residues, and cooling water. Effective treatment methods include:

- Primary treatment to remove oil and suspended solids from separator
- Biological treatment to degrade organic compounds using aerobic or anaerobic digesters.

4. FIRE AND EXPLOSION SAFETY

The Flash Point of styrene is ($\sim 31^{\circ}\text{C}$), Styrene requires strict fire safety precautions:

- Use of explosive-proof equipment and intrinsically safe electrical systems
- Inert gas blanketing (e.g. nitrogen) of storage tanks to prevent ignition from oxygen.
- Installation of gas leak detectors, automatic sprinkler systems, and fire fighting equipment throughout the plant.

5. STORAGE AND HANDLING SAFETY

Styrene must be stored and handled with great care to prevent degradation and polymerization:

- Stored in stainless steel or carbon steel tanks equipped with temperature control system to prevent overheating.
- Inhibitors are added to prevent polymerization during storage and transport.
- Pressure-relief valves, flame arrestors, and level indicators are used to maintain safe tank operations.

6. SENSITIVITY TO HEAT AND LIGHT

Styrene is thermally and photochemically unstable, all storage and process equipment must be shielded from sunlight and equipped with temperature regulation systems. Process lines and storage vessels should be insulated and, in some case jacketed for cooling. This helps maintain product quality and prevents dangerous reaction due to overheating.

7. TEMPERATURE CONTROL IN STORAGE

Styrene must be stored at temperature below 30°C to avoid polymerization and vaporization. Refrigerated or Temperature controlled tanks are recommended.

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